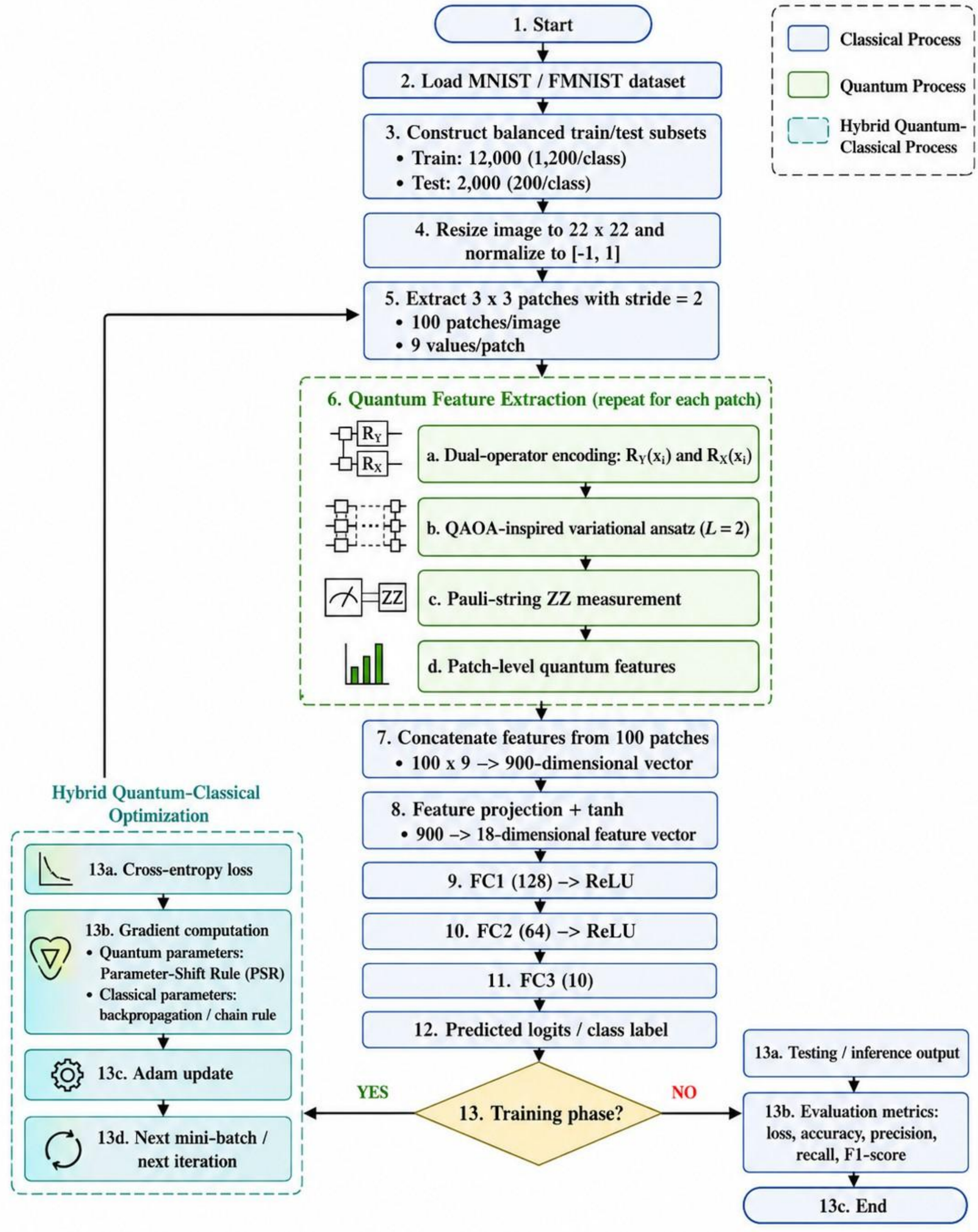


End-to-end training and inference workflow of the proposed QIFE method.

BEST PERFORMANCE OF THE METHODS ON THE MNIST DATASET (BOLD IS THE BEST VALUE)

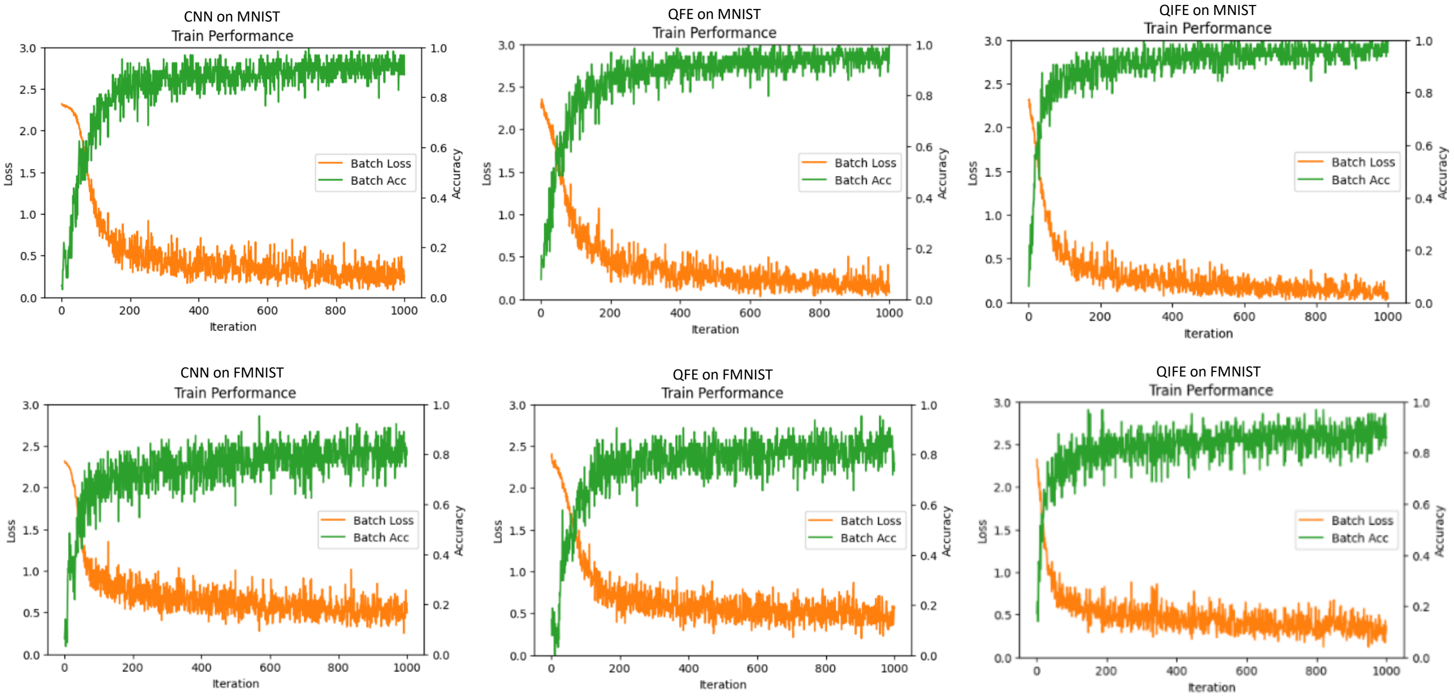
Modules and Parameters	CNN	QFE	QIFE
First Layer	Convolution 1 (1, 18, 3 × 3, stride=2, pad=1)	QFE1 (9 Qubits – 2 Layers)	QIFE (9 Qubits – 2 Layers)
Number of output channels	1 × 18 × 11 × 11	18 channels (with linear dimension projection)	18 channels (with linear dimension projection)
Pooling 1	Channel Pooling (MaxPooling2d) 1 × 9 × 11 × 11	Channel Pooling (MaxPooling1d) 9 channels	-
Second layers	Convolution 2 (9, 36, 3 × 3, stride=2, pad=1)	QFE2 (9 Qubits – 2 Layers)	-
Number of output channels	1 × 36 × 6 × 6	36 channels (with linear dimension projection)	-
Pooling 2	Channel Pooling (MaxPooling2d) 1 × 18 × 6 × 6	Channel Pooling (MaxPooling1d) 18 channels	-
Classifier	FC Layer FC1(128) – FC2(64) – FC3(10)	FC Layer FC1(128) – FC2(64) – FC3(10)	FC Layer FC1(128) – FC2(64) – FC3(10)
Batch Size	64	64	64
Optimizer	Adam	Adam	Adam
Learning Rate	0.0005	0.0005	0.0005
Iteration	1000 (5 epoch)	1000 (5 epoch)	1000 (5 epoch)
Dataset (Balanced stratified subset 10 classes)	Train = 12,000 (1,200/class); Test = 2,000 (200/class).	Train = 12,000 (1,200/class); Test = 2,000 (200/class).	Train = 12,000 (1,200/class); Test = 2,000 (200/class).
Trainable Parameters	95,110	27,988	**27,592**
FC-head elapsed time	1.03s	0.25s	**0.23s**
Test Error	0.0710	0.0640	**0.0485**
Test Acc	0.9290	0.9360	**0.9515**
Precision	0.9292	0.9362	**0.9521**
Recall	0.9290	0.9360	**0.9515**
F1-Score	0.9287	0.9359	**0.9515**

Note: FC-head elapsed time measures only the classical fully connected classifier after feature extraction; it excludes QFE/QIFE quantum-circuit execution.

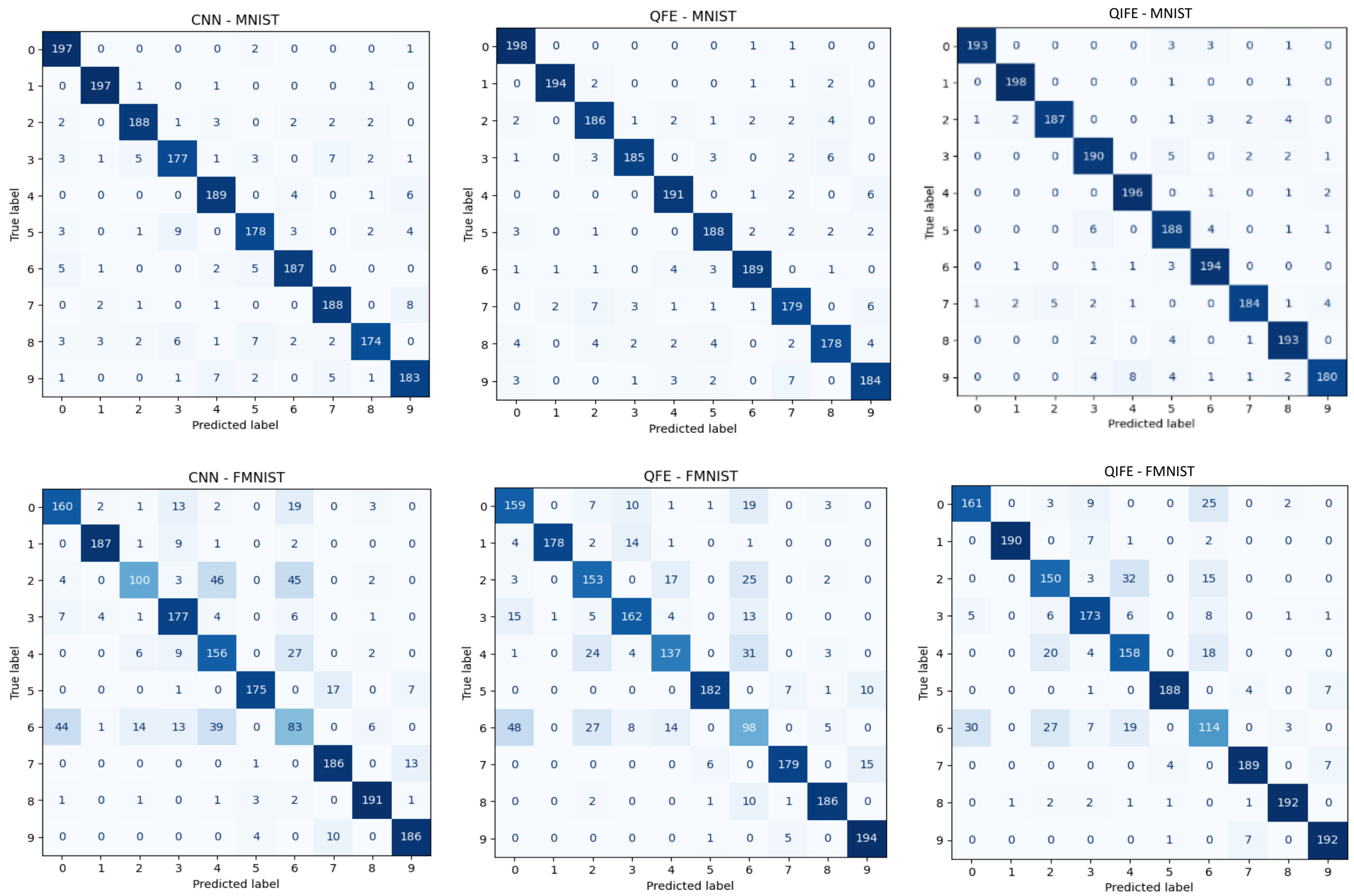
BEST PERFORMANCE OF THE METHODS ON THE FMNIST DATASET (BOLD IS THE BEST VALUE)

Modules and Parameters	CNN	QFE	QIFE
First Layer	Convolution 1 (1, 18, 3 × 3, stride=2, pad=1)	QFE1 (9 Qubits – 2 Layers)	QIFE (9 Qubits – 2 Layers)
Number of output channels	1 × 18 × 11 × 11	18 channels (with linear dimension projection)	18 channels (with linear dimension projection)
Pooling 1	Channel Pooling (MaxPooling2d) 1 × 9 × 11 × 11	Channel Pooling (MaxPooling1d) 9 channels	-
Second layers	Convolution 2 (9, 36, 3 × 3, stride=2, pad=1)	QFE2 (9 Qubits – 2 Layers)	-
Number of output channels	1 × 36 × 6 × 6	36 channels (with linear dimension projection)	-
Pooling 2	Channel Pooling (MaxPooling2d) 1x18x6x6	Channel Pooling (MaxPooling1d) 18 channels	-
Classifier	FC Layer FC1(128) – FC2(64) – FC3(10)	FC Layer FC1(128) – FC2(64) – FC3(10)	FC Layer FC1(128) – FC2(64) – FC3(10)
Batch Size	64	64	64
Optimizer	Adam	Adam	Adam
Learning Rate	0.0005	0.0005	0.0005
Iteration	1000 (5 epoch)	1000 (5 epoch)	1000 (5 epoch)
Dataset (Balanced stratified subset 10 classes)	Train = 12,000 (1,200/class); Test = 2,000 (200/class).	Train = 12,000 (1,200/class); Test = 2,000 (200/class).	Train = 12,000 (1,200/class); Test = 2,000 (200/class).
Trainable Parameters	95,110	27,988	**27,592**
FC-head elapsed time	0.90s	0.25s	**0.21s**
Test Error	0.1995	0.1860	**0.1465**
Test Acc	0.8005	0.8140	**0.8535**
Precision	0.8035	0.8185	**0.8538**
Recall	0.8005	0.8140	**0.8535**
F1-Score	0.7968	0.8148	**0.8532**

Note: FC-head elapsed time measures only the classical fully connected classifier after feature extraction; it excludes QFE/QIFE quantum-circuit execution.



Best train performance of the methods.



Best test performance of the methods.

PERFORMANCE CONSISTENCY ACROSS FIVE INDEPENDENT RUNS ON THE MNIST DATASET (**BOLD** IS THE BEST VALUE)

SETUP		PERFORMANCE							
RUN	METHODS	TRAIN LOSS	TEST LOSS	TRAIN ACC	TEST ACC	PRECISION	RECAL	F1-SCORE	FC-HEAD ELAPSED TIME
1	CNN	**0.0821**	0.2760	**0.9844**	0.9145	0.9169	0.9145	0.9138	0.86s
1	QFE	0.4049	0.3066	0.8750	0.9165	0.9172	0.9165	0.9164	**0.23s**
1	QIFE	0.1659	**0.2071**	0.9688	**0.9305**	**0.9326**	**0.9305**	**0.9306**	0.24s
2	CNN	0.2029	0.2431	0.9531	0.9260	0.9285	0.9260	0.9254	0.86s
2	QFE	0.2507	0.2579	0.9219	0.9215	0.9254	0.9215	0.9216	0.31s
2	QIFE	**0.1717**	**0.2262**	**0.9688**	**0.9295**	**0.9317**	**0.9295**	**0.9297**	**0.25s**
3	CNN	0.2537	0.2428	0.9062	0.9265	0.9278	0.9265	0.9265	0.89s
3	QFE	0.3449	0.3018	0.8750	0.9155	0.9165	0.9155	0.9152	**0.24s**
3	QIFE	**0.1157**	**0.1710**	**0.9531**	**0.9510**	**0.9518**	**0.9510**	**0.9509**	0.26s
4	CNN	0.2600	0.2371	0.8906	0.9290	0.9292	0.9290	0.9287	1.03s
4	QFE	0.0791	0.2273	**1.0000**	0.9360	0.9362	0.9360	0.9359	0.25s
4	QIFE	**0.0450**	**0.1538**	**1.0000**	**0.9515**	**0.9521**	**0.9515**	**0.9515**	**0.23s**
5	CNN	0.2504	0.2347	0.8750	0.9235	0.9250	0.9235	0.9234	0.87s
5	QFE	0.3041	0.2532	0.9219	0.9170	0.9179	0.9170	0.9166	**0.24s**
5	QIFE	**0.1423**	**0.2260**	**0.9531**	**0.9440**	**0.9464**	**0.9440**	**0.9443**	**0.24s**
		AVERAGE PERFORMANCES							
	CNN	0.2098	0.2467	0.9219	0.9239	0.9255	0.9239	0.9236	0.90s
	QFE	0.2767	0.2694	0.9188	0.9213	0.9226	0.9213	0.9211	0.25s
	QIFE	**0.1281**	**0.1968**	**0.9688**	**0.9413**	**0.9429**	**0.9413**	**0.9414**	**0.24s**

Note: FC-head elapsed time measures only the classical fully connected classifier after feature extraction; it excludes QFE/QIFE quantum-circuit execution.

PERFORMANCE CONSISTENCY ACROSS FIVE INDEPENDENT RUNS ON THE FMNIST DATASET (**BOLD** IS THE BEST VALUE)

SETUP		PERFORMANCE							
RUN	METHODS	TRAIN LOSS	TEST LOSS	TRAIN ACC	TEST ACC	PRECISION	RECAL	F1-SCORE	FC-HEAD ELAPSED TIME
1	CNN	0.3952	0.5468	**0.9062**	0.7985	0.7978	0.7985	0.7970	1.33s
1	QFE	0.5664	0.4977	0.7656	0.8140	0.8185	0.8140	0.8148	0.25s
1	QIFE	**0.3605**	**0.4463**	0.8750	**0.8350**	**0.8392**	**0.8350**	**0.8340**	**0.23s**
2	CNN	0.4942	0.5780	0.8125	0.7810	0.8148	0.7810	0.7807	0.83s
2	QFE	0.5420	0.5396	0.7812	0.7970	0.7955	0.7970	0.7941	**0.24s**
2	QIFE	**0.2541**	**0.4737**	**0.8750**	**0.8330**	**0.8358**	**0.8330**	**0.8304**	0.25s
3	CNN	0.4790	0.5876	0.7812	0.7830	0.7830	0.7830	0.7771	0.85s
3	QFE	0.5265	0.5463	0.7656	0.8055	0.8012	0.8055	0.7987	**0.25s**
3	QIFE	**0.2899**	**0.4391**	**0.8906**	**0.8340**	**0.8370**	**0.8340**	**0.8343**	**0.25s**
4	CNN	0.5000	0.5292	0.8125	0.8005	0.8035	0.8005	0.7968	0.90s
4	QFE	0.3086	0.5568	0.9219	0.8070	0.8088	0.8070	0.8035	0.25s
4	QIFE	**0.3302**	**0.3937**	**0.8594**	**0.8535**	**0.8538**	**0.8535**	**0.8532**	**0.21s**
5	CNN	0.6876	0.5921	0.7188	0.7825	0.7874	0.7825	0.7809	0.98s
5	QFE	0.6040	0.5243	0.7188	0.8130	0.8199	0.8130	0.8131	**0.25s**
5	QIFE	**0.2158**	**0.4421**	**0.9375**	**0.8355**	**0.8351**	**0.8355**	**0.8325**	0.26s
		AVERAGE PERFORMANCES							
	CNN	0.5112	0.5667	0.8062	0.7891	0.7973	0.7891	0.7865	0.98s
	QFE	0.5095	0.5329	0.7906	0.8073	0.8088	0.8073	0.8048	0.25s
	QIFE	**0.3301**	**0.4436**	**0.8844**	**0.8382**	**0.8361**	**0.8327**	**0.8311**	**0.24s**

Note: FC-head elapsed time measures only the classical fully connected classifier after feature extraction; it excludes QFE/QIFE quantum-circuit execution.

STATISTICAL SUMMARY OF PRIMARY TEST-ACCURACY RESULTS ACROSS FIVE INDEPENDENT PERFORMANCE RUNS

Datasets	Methods	Test Accuracy (%) Mean ± SD	95% CI (%)	Raw p	Holm-adjusted p vs. QIFE ($\alpha = 0.05$)	Interpretation ($p \leq$ Holm-adjusted vs. QIFE)
MNIST	CNN	92.39 ± 0.56	[91.69, 93.09]	0.018200	0.024700	Significance
	QFE	92.13 ± 0.85	[91.07, 93.19]	0.012300	0.024700	Significance
	QIFE	94.13 ± 1.07	[92.80, 95.46]	-	-	-
FMNIST	CNN	78.91 ± 0.95	[77.72, 80.10]	0.000029	0.000116	Significance
	QFE	80.73 ± 0.68	[79.88, 81.58]	0.000291	0.000872	Significance
	QIFE	83.82 ± 0.86	[82.75, 84.89]	-	-	-

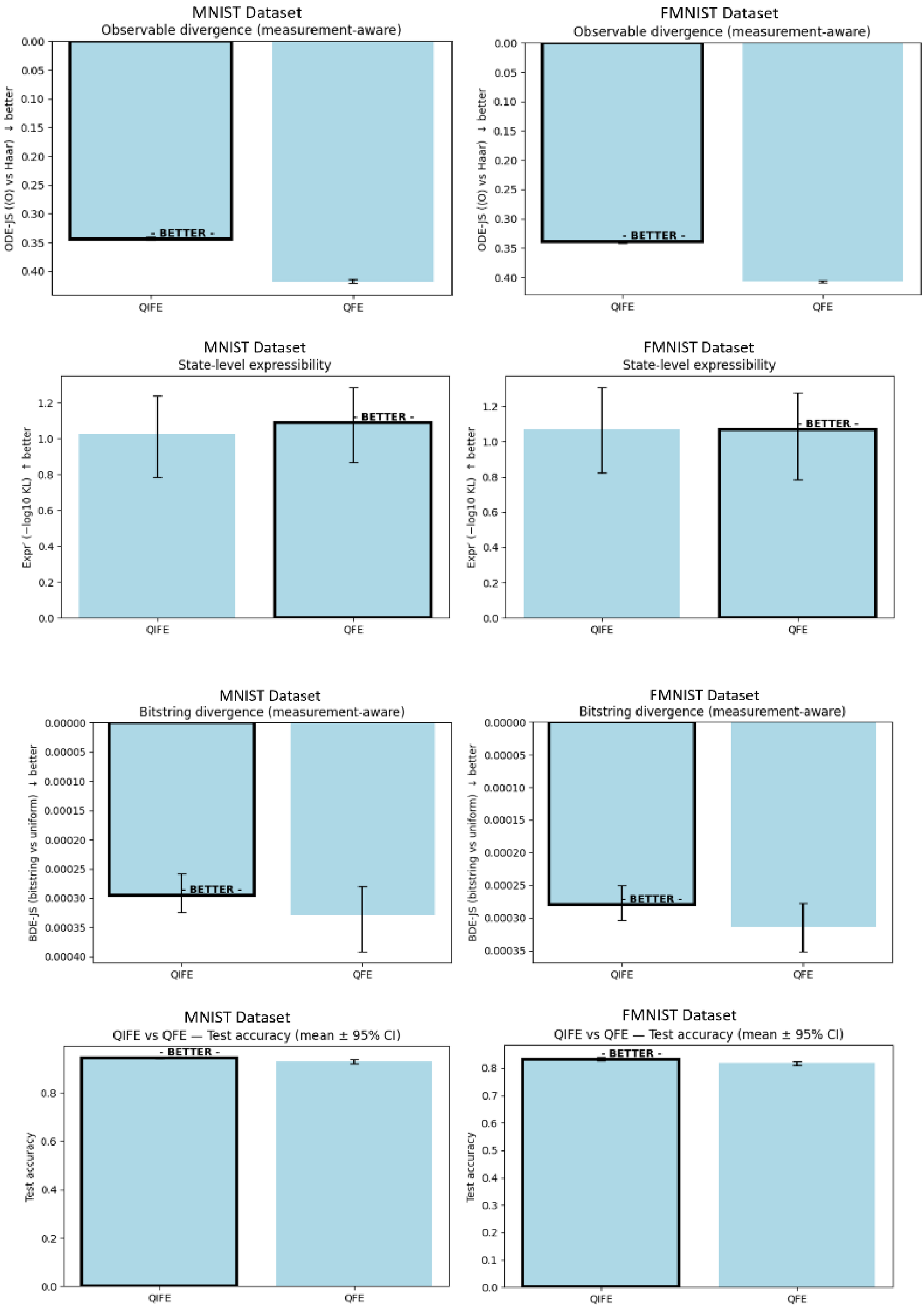
COMPARISON OF TRAINABLE PARAMETERS (**BOLD** IS THE BEST VALUE)

Methods	Dataset Dimensions	Block-1	Block-2	FC-1 (128D)	FC-2 (64D)	FC-3 (10D)	Total Parameter	Parameter Storage FP32 (MiB)	Adam Parameter-State Lower Bound FP32 (MiB)
CNN	1 × 22 × 22	(1 grayscale * 3 * 3 kernel * 18 dimensions) + 18 bias = 180	(9 dimensions * 3 * 3 kernel * 36 dimensions) + 36 bias = 2952	(648 dimensions * 128 neuron) + 128 bias = 83072	(128 * 64) + 64 = 8256	(64 * 10) + 10 = 650	95,110	0.363	1.451
QFE	1 × 22 × 22	2 Layers (9 RZ + 9 RX) + (100 patch * 9 observables * 18 dimensions) + 18 bias = 16254	2 Layers (9 RZ + 9 RX) + (9 observables * 36 dimensions) + 36 bias = 396	(18 dimensions * 128 neuron) + 128 bias = 2432	(128 * 64) + 64 = 8256	(64 * 10) + 10 = 650	27,988	0.107	0.427
QIFE	1 × 22 × 22	2 Layers (9 RZ + 9 RX) + (100 patch * 9 observables * 18 dimensions) + 18 bias = 16254	-	(18 dimensions * 128 neuron) + 128 bias = 2432	(128 * 64) + 64 = 8256	(64 * 10) + 10 = 650	**27,592**	**0.105**	**0.421**

The inference parameter-storage estimate is calculated as $M_{param} = P \times b$, where P is the number of trainable parameters and $b = 4$ bytes for FP32 storage. The Adam parameter-state lower bound is approximated as $M_{Adam} \approx 4Pb$, accounting for parameters, gradients, and the first- and second-moment buffers. These analytical estimates exclude activation tensors, input batches, framework overhead, and quantum-simulator state memory.

SUMMARY OF QIFE AND QFE EXPRESSIBILITY MEASUREMENT RESULTS (**BOLD** IS THE BEST VALUE)

Measurements	MNIST QIFE	MNIST QFE	FMNIST QIFE	FMNIST QFE
Mean test acc	**0.9440**	0.9292	**0.8332**	0.8175
ODE-JS	**0.3439**	0.4174	**0.3394**	0.4069
BDE-JS	**0.00030**	0.00033	**0.00028**	0.00031
State-level	1.026	**1.090**	1.068	**1.071**
Efficient $\frac{1-ODE}{Cost_{2Q}}$	**0.0182**	0.0081	**0.0183**	0.0082
Efficient $\frac{1-BDE}{Cost_{2Q}}$	**0.0278**	0.0139	**0.0278**	0.0139
Efficient $\frac{State-level}{Cost_{2Q}}$	**0.0285**	0.0151	**0.0297**	0.0149
Cohen's	$d = 1.76 \geq 0.8$		$d = 1.63 \geq 0.8$	


MNIST Dataset
Observable divergence (measurement-aware)
ODE-JS (⟨O⟩ vs Haar) ↓ better
- BETTER -
QIFE
QFE
FMNIST Dataset
Observable divergence (measurement-aware)
ODE-JS (⟨O⟩ vs Haar) ↓ better
- BETTER -
QIFE
QFE
MNIST Dataset
State-level expressibility
Expr (−log10 KL) ↑ better
- BETTER -
QIFE
QFE
FMNIST Dataset
State-level expressibility
Expr (−log10 KL) ↑ better
- BETTER -
QIFE
QFE
MNIST Dataset
Bitstring divergence (measurement-aware)
BDE-JS (bitstring vs uniform) ↓ better
- BETTER -
QIFE
QFE
FMNIST Dataset
Bitstring divergence (measurement-aware)
BDE-JS (bitstring vs uniform) ↓ better
- BETTER -
QIFE
QFE
MNIST Dataset
QIFE vs QFE — Test accuracy (mean ± 95% CI)
Test accuracy
- BETTER -
QIFE
QFE
FMNIST Dataset
QIFE vs QFE — Test accuracy (mean ± 95% CI)
Test accuracy
- BETTER -
QIFE
QFE